Week 2

1.Which HDFS component is primarily responsible for monitoring replica counts and initiating re-replication when a block becomes under-replicated?

A. DataNode
B. Client
C. NameNode
D. Secondary NameNode

Answer: C. NameNode

Explanation

A. DataNode (Incorrect): DataNodes store actual block data and report block information to the NameNode, but they do not decide replication policies.

B. Client (Incorrect): Clients read and write files in HDFS but are not responsible for managing block replication.

C. NameNode (Correct): The NameNode maintains metadata about all blocks and replicas, monitors replication levels, and initiates re-replication when blocks become under-replicated.

D. Secondary NameNode (Incorrect): The Secondary NameNode periodically merges FsImage and EditLogs; it does not manage replica counts or perform re-replication.

2.What is the primary purpose of introducing the HDFS Federation?

A. Increase replication factor
B. Increase namespace scalability
C. Reduce block size
D. Eliminate DataNodes

Answer: B. Increase namespace scalability

A. Increase replication factor (Incorrect): Replication factor is configured separately and is not the purpose of Federation.

B. Increase namespace scalability (Correct): HDFS Federation allows multiple independent NameNodes to manage separate namespaces, removing the namespace bottleneck and improving scalability.

C. Reduce block size (Incorrect): Federation does not affect HDFS block size; block size is a storage configuration parameter.

D. Eliminate DataNodes (Incorrect): DataNodes are still required in HDFS Federation and can serve blocks for multiple namespaces.

3.Which of the following is directly affected by the number of HDFS blocks?

A. NameNode memory usage
B. Number of Map tasks
C. Both A and B
D. Neither A nor B

Answer: C. Both A and B

Explanation

C. Both A and B (Correct): Each HDFS block requires metadata storage in the NameNode, and typically each block is processed by one Map task, so both are directly affected.

A. NameNode memory usage (Correct statement): More blocks mean more metadata entries, increasing NameNode memory consumption.

B. Number of Map tasks (Correct statement): In MapReduce, the number of input splits (often corresponding to blocks) influences the number of Map tasks.

D. Neither A nor B (Incorrect): Both NameNode memory usage and Map task count are directly influenced by the number of HDFS blocks.

4.What is the default value of the parameter dfs.replication?

A. 1
B. 2
C. 3
D. 4

Answer: C. 3

Explanation

A. 1 (Incorrect): A replication factor of 1 provides no redundancy and is not the default setting.

B. 2 (Incorrect): While possible to configure, 2 is not the default value in HDFS.

C. 3 (Correct): By default, HDFS stores three replicas of each block to provide fault tolerance and data availability.

D. 4 (Incorrect): HDFS allows a replication factor of 4, but the default is 3 to balance reliability and storage overhead.

5. If a DataNode stops sending heartbeats, what does the NameNode do?

A. Ignores the failure
B. Deletes all blocks immediately
C. Marks the DataNode and initiates re-replication if needed
D. Restarts the DataNode automatically

Answer: C. Marks the DataNode and initiates re-replication if needed

Explanation

A. Ignores the failure (Incorrect): The NameNode continuously monitors heartbeats and does not ignore missing DataNodes.

B. Deletes all blocks immediately (Incorrect): The blocks are not deleted; the NameNode marks their replicas on the failed DataNode as unavailable and creates new replicas elsewhere.

C. Marks the DataNode and initiates re-replication if needed (Correct): If heartbeats stop, the NameNode considers the DataNode dead and starts re-replicating blocks to maintain the desired replication factor.

D. Restarts the DataNode automatically (Incorrect): The NameNode cannot restart a failed DataNode; recovery must be handled externally by administrators or cluster management tools.

6. Which statement about Reduce tasks is TRUE?

A. A key can be processed by multiple Reducers.
B. Each key is assigned to exactly one Reducer.
C. Reducers execute before Mappers.
D. Reducers never use partitioning.

Answer: B. Each key is assigned to exactly one Reducer

Explanation

A. A key can be processed by multiple Reducers (Incorrect): A single key and all its values must go to one Reducer to produce a correct aggregated result.

B. Each key is assigned to exactly one Reducer (Correct): The partitioner ensures that all values associated with a particular key are sent to the same Reducer for processing.

C. Reducers execute before Mappers (Incorrect): Reducers start only after Mapper output is generated and shuffled.

D. Reducers never use partitioning (Incorrect): Partitioning is essential for determining which Reducer receives each key-value pair.

7. What is the primary role of a Partitioner in MapReduce?

A. Sort records within a Reducer
B. Decide which Reducer receives a key-value pair
C. Combine Mapper outputs locally
D. Replicate data blocks across HDFS

Answer: B. Decide which Reducer receives a key-value pair

Explanation

A. Sort records within a Reducer (Incorrect): Sorting is handled during the shuffle and sort phase, not by the Partitioner.

B. Decide which Reducer receives a key-value pair (Correct): The Partitioner determines the Reducer to which each key-value pair is sent, ensuring that all values of the same key go to the same Reducer.

C. Combine Mapper outputs locally (Incorrect): This is the role of a Combiner, not a Partitioner.

D. Replicate data blocks across HDFS (Incorrect): Block replication is managed by HDFS, not MapReduce partitioning.

8. Which YARN component is responsible for detecting task failures for a specific job?

A. Resource Manager (RM)
B. Node Manager (NM)
C. Application Master (AM)
D. DataNode

Answer: C. Application Master (AM)

Explanation

A. Resource Manager (RM) (Incorrect): Resource Manager allocates cluster resources but does not monitor individual task failures for a job.

B. Node Manager (NM) (Incorrect): Node Manager manages containers on a node and reports node status, but it does not handle application-level failure recovery.

C. Application Master (AM) (Correct): The Application Master monitors the tasks of a specific application, detects task failures, and requests re-execution when necessary.

D. DataNode (Incorrect): DataNode is an HDFS storage component and is unrelated to YARN task monitoring.

9. Which statement about the Reduce function is TRUE?

A. It accepts an intermediate key and a set of values.
B. It processes only one value at a time.
C. It cannot aggregate values.
D. It always produces multiple output values.

Answer: A. It accepts an intermediate key and a set of values.

Explanation

A. It accepts an intermediate key and a set of values (Correct): The Reduce function receives an intermediate key along with all values associated with that key and processes them together.

B. It processes only one value at a time (Incorrect): A Reducer processes a key and its entire collection of values, not individual values separately.

C. It cannot aggregate values (Incorrect): Aggregation (e.g., sum, count, average) is one of the primary purposes of the Reduce function.

D. It always produces multiple output values (Incorrect): A Reducer may produce one, multiple, or even zero output records depending on the application logic.

10. In the Reverse Web-Link Graph application, the Mapper emits:

A. (source, target)
B. (target, source)
C. (URL, 1)
D. (target, count)

Answer: B. (target, source)

Explanation

A. (source, target) (Incorrect): This represents the original web-link graph and does not reverse the links.

B. (target, source) (Correct): To reverse the web-link graph, the Mapper swaps the original link direction and emits (target, source) so that all pages linking to a target can be grouped together.

C. (URL, 1) (Incorrect): This format is typically used in counting applications such as Word Count, not Reverse Web-Link Graph.

D. (target, count) (Incorrect): The Mapper emits source-target relationships, not counts; counting may occur later if required.